

RANDOM VARIABLES & BINOMIAL

new 2.8–2.10 • old 4.7–4.12 • ★ geometric + combining

ZONE A · THE FORMULAS

ANY DISCRETE RANDOM VARIABLE

$$\mu_X = \sum x_i \cdot P(x_i)$$

$$\sigma_X = \sqrt{\sum (x_i - \mu_X)^2 \cdot P(x_i)}$$

Weight each possible value by its probability.
All probabilities are nonnegative and sum to 1.

SHIFT + SCALE: $Y = a + bX$

$$\begin{aligned}\mu_Y &= a + b\mu_X \\ \sigma_Y &= |b|\sigma_X\end{aligned}$$

BINOMIAL: COUNT THE SUCCESSES

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

$$\begin{aligned}\mu_X &= np \\ \sigma_X &= \sqrt{np(1 - p)}\end{aligned}$$

Binary: success or failure
Independent trials
Number of trials is fixed
Same success probability p

Possible counts: $x = 0, 1, \dots, n$.
Check **BINS** before using the formulas.

ZONE B · WHAT THE SYMBOLS MEAN

X : a numerical outcome of a chance process
 x_i : one possible value
 μ_X : long-run mean σ_X : SD

Binomial X : successes in n trials
 p : success probability on one trial
 $\binom{n}{x}$: number of success/failure orderings

ZONE C · SAY IT IN WORDS

“If we repeated this many times, the average _____ would be about _____.”

Name the variable, the value of μ_X , and its units.

ZONE D · WATCH OUT + BEYOND

COMBINE VARIANCES

For independent variables, add **variances**; then take the square root. Do not add SDs.
Subtracting variables still adds their variances.
Use $|b|$, not b , when scaling an SD.

★ BEYOND THE EXAM

Combining random variables

$\mu_{aX+bY} = a\mu_X + b\mu_Y$
 $\text{Var}(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2$
Mean rule: always. Variance rule: **independent** X, Y .

Geometric: trial of the first success

Independent binary trials, same $0 < p \leq 1$; count $x = 1, 2, 3, \dots$
 $P(X = x) = (1 - p)^{x-1} \cdot p$
 $\mu_X = \frac{1}{p} \quad \sigma_X = \frac{\sqrt{1 - p}}{p}$